Day-3 Sep-26

OSI Model & TCP/IP Suite

What is a networking model?

↳ Networking models categorize and provide a structure for networking protocols and standards.

Protocols: A set of rules defining how network devices and software should work including how they should work together.

OSI Model (Open Systems Interconnection)

Though OSI Model isn't actually ~~isn't~~ in use today, it has had a big impact on how network engineers think about networking and we still refer it to it even today.

OSI Model Stands for "Open Systems Interconnection"!

Open – open standard (open source), not a proprietary model developed and used by individual company.

Application	Layer	(7
Presentation	Layer	(6
Session	Layer	(5
Transport	Layer	(4
Network	Layer	(3
Data Link	Layer	(2
Physical	Layer	(1

- A conceptual model that categorizes and standards standardizes the different functions in a network
- Created by ISO (International Organization for Standardization)
- Functions are divided into 7 'layers'.
- These layers work together to make the network work

Sender - Original data मा each layer .मा कोहि न कोहि extra header चाहिन्छ which is called encapsulation.

de-encapsulation Receiver को मा चाहिन्छ every layer मा रहेको sender को corresponding layer मा add गरेको header remove हुनेछ which is called decapsulation.

7) Layer 7 - Application layer.

- This layer is closest to the end user
- It interacts with software application that have some communication component., such as web browsers (chrome, brave, Firefox etc)
- Eg: HTTP and HTTPS are layer 7 का protocols. ex https://www.cisco.net

Functions of layer 7

- Identifying communication partners
- Synchronizing communication.

The communication between the application layers of two different systems is called same-layer interaction.
This allows application layer to perform its functions.

Layer 6) Presentation layer.

- Data in the application layer is in "application format".
- It needs to be "translated" to a different format to be sent over the network.
- Presentation layer's job is to translate between application & network formats.
- For eg, encryption of data as it is sent, and decryption of data as it is received.
- Also translates between different Application layer formats.

(Presentation layer translates data to appropriate format)

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Layer 5 Session layer.

- Controls dialouges (sessions) between communicating hosts.
- Establishes, manages and termincites connection between local application (eg. our browser) and remote application (for eg, Youtube).

NOTE: The Upper Layers (Layer 7, 6, 5)

- Application, Presentation, Session layers are ~~called~~ upper layer.
- Network engge engineers usually don't work coith top 3 layers.
- Application developers work with top layers of OSI model to connect their applications over network.

Layer-4 Transport Layer

- It segments (breaks into small pieces) and reassembles data for communication between end hosts.
- Breaks large piece of data into smaller segments which can be sent more easily sent over the network and are likely to cause transmission problems if occur.

- Provide host-to-host communication

(So, top 3 layers or upper layer ले data आउँछ, transport layer ले उ त्यो data लाई small pieces called segment मा break gariinxa. Ani every segment मा transport layer ले header join/add gardera network layer मा पठाउँछ.

Layer-3 Network layer.

- Provides connectivity between end hosts on different networks (i.e outside of LAN).
- Provides logical addressing (IP addresses).
- Provides path selection between source and destination
- Routers operate at layer 3.

(यो ले Layer ले segment लिन्छ from transport layer. Segment मा ले layer 3 को header include गरिन्छ, which includes source & destination IP address). Now it is called packet. Packet = data + layer 4 header + layer 3 header.

Layer 2 Data Link layer.

- Provides node-to-node connectivity and data transfer (for e.g, PC to ⌀ switch, switch to router, router-to-router)
- Defines how data is formatted for transmission over a physical medium (for. eg, copper UTP cables).
- Detects and (possibly) corrects Physical Layer error.
- Uses layer 2 addressing, separate from layer 3 addressing.
- Switches operate at level 2.

(Packet लिन्छ) layer 2 ले (अगाडि) layer 2 header र layer 2 trailer हाल्छ। Now it is called frame).
(पछि layer ले MAC address add गरिदिन्छ।)

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Layer-1 Physical layer.

- Defines physical characteristics of the medium used to transfer data between devices.

- For example; voltage levels, max^m transmission distances, physical connectors, cable specifications etc.

- Digital bits are converted into electrical (for wired connections) or radio (for wireless connections) signal.

- All of the info in day's video (cables, pin, layout, etc.) is related to physical layer)

PDUs (Protocol data units)

Data		Data	Top 3 layer
Data + L4 Header	→	Segment	Layer 4
Data + L4 header + L3 header	→	Packet	Layer 3
L2 trailer + Data + L4 header + L3 header	→	Frame	Layer-2
		Bit	layer-1

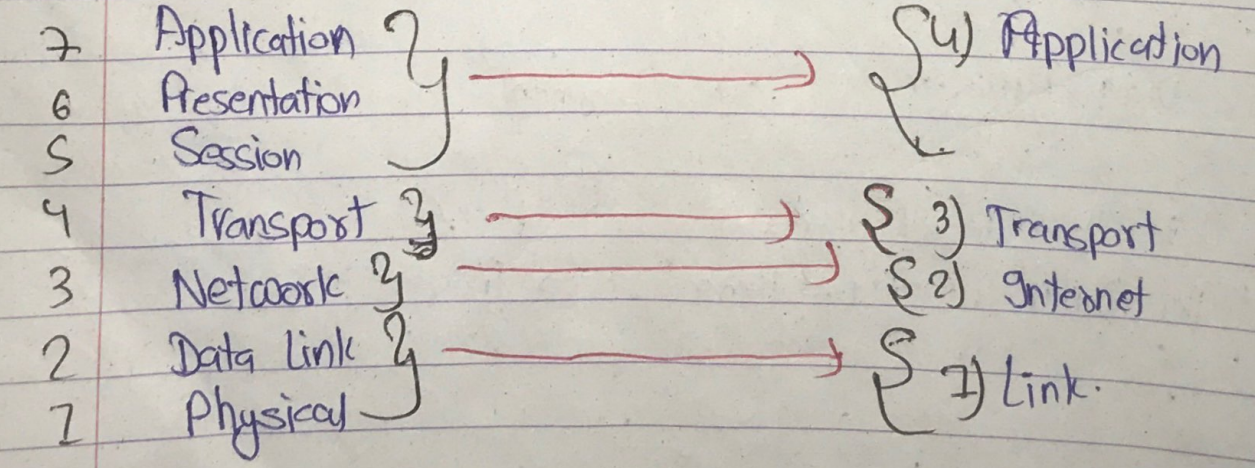
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OSFP - Finds shortest path, routing here.

TCP/IP Suite.

- Conceptual model and set of communication protocols used in Internet and other networks.
- Known as TCP/IP because these are two of foundational protocols in the suite.
- Developed by US Department of Defense.
- Similar structure to OSI model, but with fewer layers.
- This is the model actually in use in modern networks

NOTE: The OSI model still influences how network engineer think and talk about networks



When talking about networks, we use the OSI numbering. For eg- If you say there is layer 4 problem in the network, network engineer will think of OSI layer's 4 i-e transport layer not TCP/IP layer 4 Application.

STP - Spanning Tree Protocol (Layer-2 Protocol) has layer 2 & layer 1 information

OSPF - Layer 3 protocol. has information about layer 3, layer 2 & layer 1
↳ Open Shortest Path First

IP address - Layer 3 information.

Supplementary Material

Q.1) HTTP data sent from YT web surfer is displayed via your web browser. This is an example of what.
a) Adjacent-layer interaction (adjacent - different layers)
b) Same-layer interaction
c) Encapsulation.
d) De-encapsulation.

Q.2) HTTP data has been encapsulated with three separate headers and one trailer. What is the appropriate name for this PDU
a) Packet b) Segment c) Frame d) Data

Q.3) Which layers of OSI Model are most-relevant to the role of a network engineer?
a) transport - network - data link physical
b) transport - network - datalink.
c) network only
d) Application - Presentation - Session

Q.4) The link layer of TCP/IP model is equivalent to what layer, or layers, of OSI model.
a) Transport - network
b) Network - data link
c) Data link
d) Data + link - Physical

Q.5) Which layer of OSI provides host-to-host communication
a) Application b) Network c) Transport d) Data-Link

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